Sam Toughan

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About Me

First-Class Computer Science graduate, University of Bristol; highest-ranked software engineering dissertation in the cohort (4th overall of 250). ML engineer with production deployment experience, undergraduate Principal Investigator on NHS-linked research, and creator of Acquisition, a live language-learning application with a custom natural language generation engine.

Education and Qualifications

Sep 22 – Jun 26 **University of Bristol | BSc Computer Science, First-Class Honours**

- Dissertation: 83% (highest software engineering dissertation; 4th overall of 250).
- 1st in cohort in Image Processing & Computer Vision with 90%
- Ranked 17th of 200 (75% average) in final year.

Sep 19 – Jun 21 **Sixth Form (A-Levels)**

- Maths A*, Computer Science A* and Physics A.

Work Experience

Feb 26 – Present **University of Bristol, in collaboration with the NHS**
Principal Investigator

- Building machine learning models to simulate responses to urban interventions using environmental and mobility data.
- Integrating green space, mobility, health, and carbon datasets for climate and wellbeing analysis.
- Designing a metric to quantify the combined climate and health impact of urban interventions.

May 26 – Jul 26 **Axeria**
ML Engineer

- Fine-tuned multi-task transformer classifiers for a production LLM-routing system on H200 nodes.
- Built the model acceptance evaluation pipeline and deployed models to NVIDIA Triton (ONNX→TensorRT).
- Integrated the routing engine into an enterprise LiteLLM gateway with a secure tunnel for self-hosted models.

Aug 25 – Jun 26 **University of Bristol**
Graduate Teacher

- Delivered first-year seminars in functional programming, imperative programming, and algorithms.
- Ran Computer Systems labs in C and operating systems, grading coursework and giving consistent feedback.

Jun 25 – Aug 25 **Vodafone**
AI & Automation Engineer Intern

- Automated weekly and monthly reporting, reducing preparation time from 2 hours to 30 minutes with validation checks.
- Built ETL pipelines from internal systems and APIs, adding anomaly flags and owner notifications.
- Introduced monitoring and runbooks to improve automation reliability and operational support.

May 25 – Jun 25 **University of Bristol**
Software Engineer Intern

- Co-built an interactive programming game for Bristol CS Open Days using JavaScript-based controls.
- Built the React frontend, browser-based parser, and PostgreSQL-backed APIs, then deployed the platform to the cloud.
- Used by 300+ visitors, with telemetry-driven changes reducing level completion errors by 25%.

Dec 24 – Mar 25 **IBM**
Software Engineer (Contract)

- Built a cross-platform weather application combining live forecasts, voice interaction, and generative visual output.
- Developed Flutter frontend features and backend services in Flask and MongoDB, integrating external APIs for weather.
- Improved system robustness through error handling, fallback flows, containerised deployment, and CI/CD automation.

Projects

Acquisition Spanish Learning Web App (Next.js, React, TypeScript, Supabase, Python)

- Built and deployed Acquisition (languageacquisition.net), a Spanish learning PWA with flashcards, reading, and listening.
- Engineered a custom Kneser-Ney NLG pipeline with target-anchored beam search over a 95M-sentence-corpus.
- Designed a custom heuristic spaced repetition scheduler with five outcome paths and adaptive workload sizing.
- Evaluated the platform in a two-week dissertation study; 9 of 10 participants in the primary analysis improved by at least one measured receptive CEFR band

Lattice Boltzmann Solver Optimisation (C, OpenMP, MPI, CUDA, HPC profiling)

- Optimised a D2Q9-BGK lattice Boltzmann solver in C on the Isambard-AI supercomputer.
- Added OpenMP, MPI domain decomposition, and CUDA variants across CPU and GPU hardware.
- Improved runtime by 197x through memory layout changes, vectorised kernels, roofline analysis, and profiling.

Cornell Box Rendering Project (C++ / OOP / Rasterising / Ray Tracing)

- Built a Cornell Box renderer from first principles in C++ using object-oriented design.
- Implemented physically based materials, environment mapping, depth of field, and camera models.

Skills:

Programming: C, C++, Python, Go, Java, Haskell, SQL, TypeScript

Frameworks & Tools: React, Next.js, Flutter, Flask, Supabase, PostgreSQL, Docker, Git, CI/CD

Interests:

Water polo (inter-university competitions), marathon running, guitar performances, language learning (Spanish, French and Italian), and Vice President of the University of Bristol Spanish Society.

Achievements:

Appointed Principal Investigator on a University of Bristol research project as an undergraduate student.

Awarded the Wertheimer Memorial Prize for the best contribution to the Faculty and fellow students, University of Bristol.

Reached C2 Spanish within one year of self-study.

Co-founded the First Uni of Bristol European Formal Ball with 200 attendees as VP of the Spanish Society.

Languages:

English (C2), Arabic (C2), Spanish (C2), French (B1), Italian (A2)